ISEA • CRYPTOGRAPHY

ZKP Based Multi-System Attribute Verification

Lecture 1: Motivation & Problem Formulation

Debrup Chatterjee • Hackathon 2026 • ISI Kolkata × Fortytwo Labs

Company Overview

"Fortytwo Labs is a deep-tech cybersecurity platform company building quantum-safe digital trust infrastructure for regulated and mission-critical environments.

Through its π-Control platform, the company secures identities, APIs, applications, and sensitive data using patented, indigenous cryptographic innovations designed for long-term resilience.

Focused on Banking, FinTech, Defence, and Strategic Government sectors, Fortytwo Labs enables organizations to move from fragmented security tools to a unified, future-ready trust foundation."

Copyright © Fortytwo Labs

FORTYTWO
SECURE TODAY. FUTURE READY.

Awards & Recognition

Top 5 Investible Companies Award
ESTIC 2025

SIDM Defense Champion Award 2023

Best Quantum-Safe Cybersecurity Company: CISO Conclave Awards 2025

NASSCOM Emerge 50
League of 10 Winner 2024

Bharat Cybersurakhsha Summit Emerging Technology Award 2025

CIO TECH OUTLOOK 10 MOST PROMISING DIGITAL & IDENTITY AUTHENTICATION TECHNOLOGY SOLUTIONS STARTUPS 2025

FINANCE OUTLOOK INDIA 10 MOST PROMISING FINTECH COMPANIES - 2024

Best Tech Startup of the year 2023
TECHINDIA Transformation Awards

Best in Cybersecurity Innovation 2023
MSME INDIA BUSINESS AWARDS

Industry Outlook

TOP 10 DEFENCE STARTUPS 2022

Copyright © Fortytwo Labs

FORTYTWO
SECURE TODAY. FUTURE READY.

"What if you could prove who you are... without revealing who you are?"

The identity paradox of the digital age

FORTYTWO
SECURE TODAY. FUTURE READY.

Meet Priya - Our Running Example

IDENTITY ATTRIBUTES

Age:	**24 years**
Annual Income:	**₹12 LPA**
Employer:	**TechCorp India**
Education:	**B.Tech (CS) + M.Tech (Cybersecurity)**
Health Status:	**Non-smoker, No pre-existing conditions**
Address:	**Kolkata, West Bengal**
Aadhaar:	**XXXX-XXXX-1234**
PAN:	**ABCDE1234F**

FORTYTWO
SECURE TODAY. FUTURE READY.

One Day in Priya's Digital Life

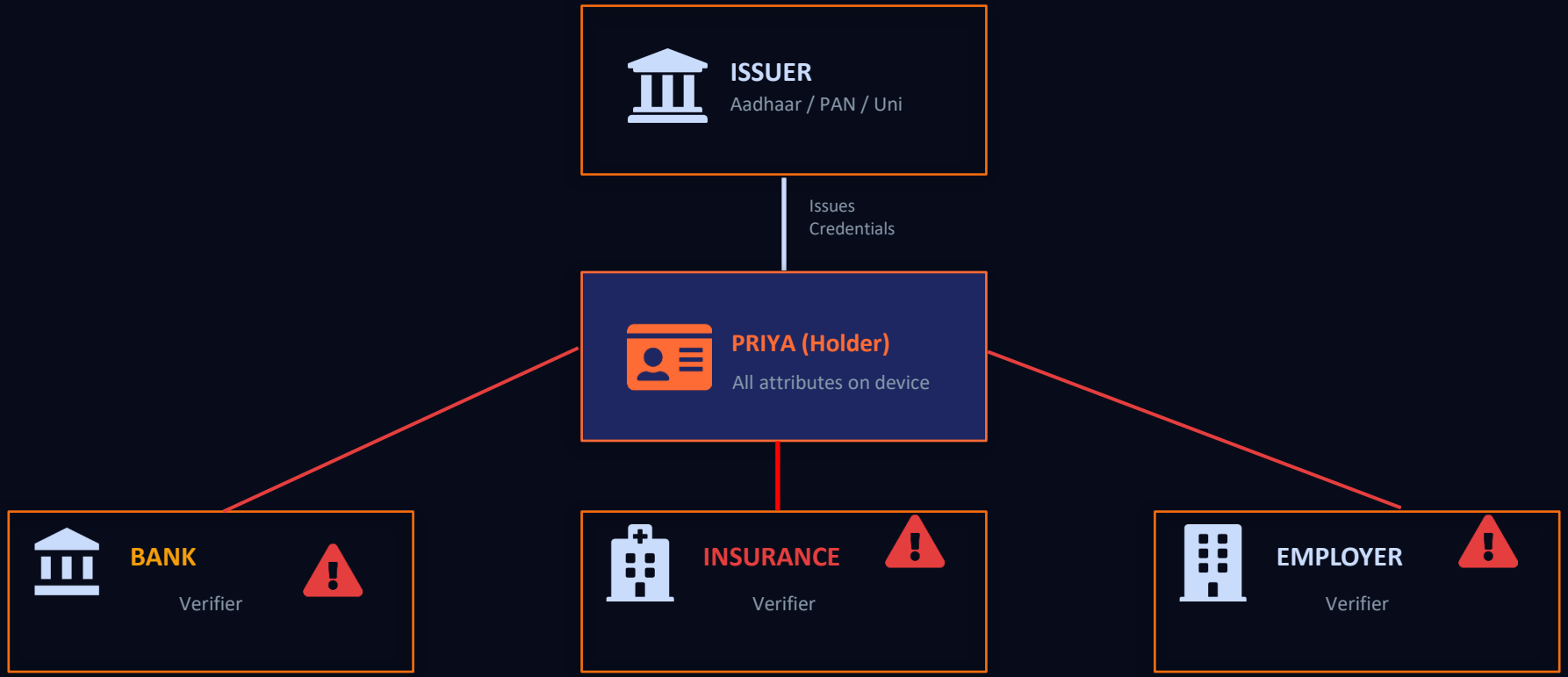
Each service asks for MORE than it needs

Morning	Afternoon	Evening	Night
Bank Loan	**Health Insur.**	**Job Portal**	**Age Verif.**
ASKS FOR:	ASKS FOR:	ASKS FOR:	ASKS FOR:
Show full income proof + ID	Share complete medical history	Verify employment + qualifications	Prove age > 18

Privacy Impact: Every verification exposes MORE data than necessary - creating a trail across systems.

FORTYTWO
SECURE TODAY. FUTURE READY.

Current State: The Centralized Identity Model



presents FULL data to ALL verifiers

FORTYTWO
SECURE TODAY. FUTURE READY.

Component 1: The Issuer

ISSUER

DEFINITION

An authority that attests to and vouches for the truth of a holder's attributes by signing them cryptographically.

KEY PROPERTY: TRUSTED authority whose attestation carries cryptographic weight

REAL-WORLD EXAMPLES

Government

Aadhaar, PAN Card, Passport

University

Degree certificates, transcripts

Employer

Employment proof, salary slips

Bank

Income statements, KYC

FORTYTWO
SECURE TODAY. FUTURE READY.

Component 2: The Holder (User)

HOLDER

DEFINITION

The individual who possesses identity attributes and must prove specific claims about them to various verifying parties.

KEY CHALLENGES PRIYA FACES

- Must share MORE data than the verifier actually needs
- Cannot control what the verifier does with her data
- Same sensitive data repeated to EVERY verifier
- No technical guarantee against data misuse or re-sharing

FORTYTWO
SECURE TODAY. FUTURE READY.

Component 3: The Verifiers

Entities that need to verify specific claims about the holder - but each only NEEDS specific information, yet receives everything.

THE CATCH: Each verifier only NEEDS specific information... but they GET everything. Here's the disparity:

VERIFIER	WHAT THEY NEED	WHAT THEY GET
Bank	Income > ₹5 LPA (1 bit: YES/NO)	Full salary slips + bank statements + PAN + transaction history
Insurance Co.	Age ≥ 18 + Non-smoker (2 bits)	Complete medical records + full health history + family conditions
Job Portal	Has M.Tech degree (1 bit: YES/NO)	Entire academic transcript + grades + institute + faculty references

Data Needed vs. Data Shared Ratio - typically 1:100 or worse. This is the core problem ZKP solves.

FORTYTWO
SECURE TODAY. FUTURE READY.

The Information Asymmetry Problem

WHAT VERIFIER NEEDS

"Is Priya's income above ₹5 Lakhs?"

= 1 bit of information

VS

WHAT VERIFIER ACTUALLY GETS

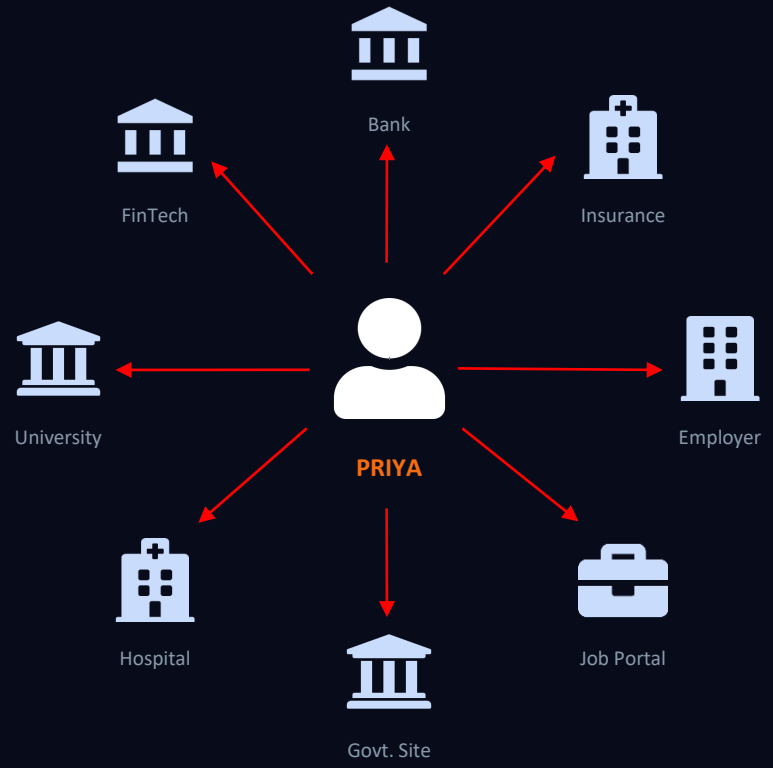
≈ 19 MB of sensitive data

"The system reveals everything when it only needs to reveal a single fact."

FORTYTWO
SECURE TODAY. FUTURE READY.

The Multi-Verifier Nightmare

Priya interacts with 8 services in a month - every interaction leaves a data trail



PRIVACY RISKS

- Build detailed profile
- Correlate with other data
- Share or sell data
- Link activities cross-system
- Infer hidden attributes

⚠ LINKED!

FORTYTWO
SECURE TODAY. FUTURE READY.

Your Hackathon Challenge

OFFICIAL PROBLEM STATEMENT

Design and build a zero-knowledge proof (ZKP) system that allows a user to prove selected attributes (or predicates over attributes) to multiple independent verifiers across systems without revealing the underlying attributes and while preventing linkage across verifications.

Selective Disclosure

Prove selected attributes or predicates over attributes - not all attributes at once

Multi-Verifier

Work with MULTIPLE independent verifiers across different systems simultaneously

Zero Disclosure

WITHOUT revealing the underlying attribute values to any verifier

Unlinkability

PREVENTING linkage across verifications - verifiers cannot correlate proofs

🏆 **Prize Pool: ₹2,00,000 | Duration: Jan 27 – Mar 10, 2026 | Teams: 2–4 members**

FORTYTWO
SECURE TODAY. FUTURE READY.

Summary: Four Core Problems to Solve

01 OVER-DISCLOSURE

Systems share far more information than any verifier actually needs, exposing sensitive data unnecessarily with every interaction.

~100× more data shared than needed

02 LINKABILITY

Using the same credential across services allows verifiers - or eavesdroppers - to correlate Priya's activity across systems.

Identity traces persist across sessions

03 DATA ACCUMULATION

Each verifier builds a detailed profile over time. Individually harmless data points combine into comprehensive surveillance.

Profile completeness grows with each use

04 LOSS OF CONTROL

Once data leaves the holder's device, they have no technical means to control or audit how verifiers store, share, or use it.

No technical enforcement post-disclosure

Next: Can conventional identity systems solve these problems? Let's find out →

SECTION 2 - CONVENTIONAL APPROACHES

Why Conventional Techniques Fail

2

"We already have identity solutions... or do we?"

Slides 13–22

FORTYTWO
SECURE TODAY. FUTURE READY.

Approach 1: Direct Document Sharing

User
Has identity documents

→

Scans
Creates digital copies

→

Sends
Uploads to verifier

→

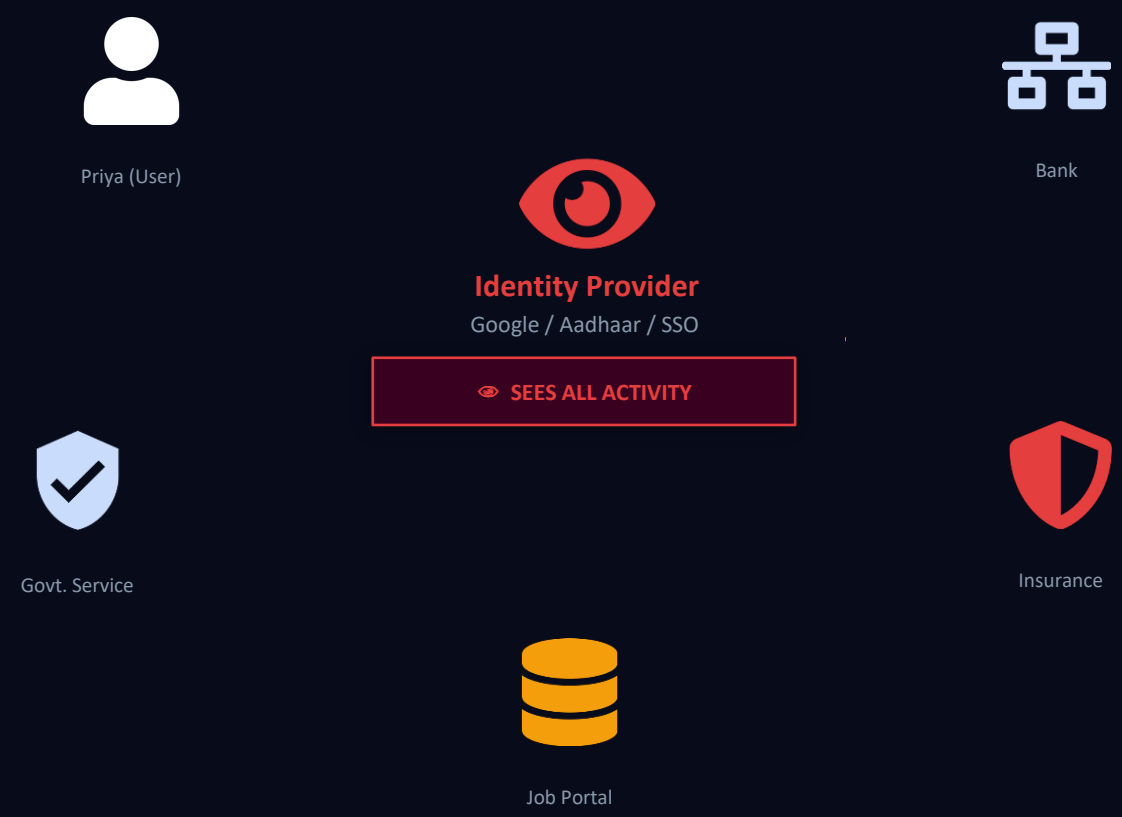
Verifier
Receives & stores all data

PROBLEMS WITH THIS APPROACH

- ✗ Full data exposure - every field, every document is shared
- ✗ No attribute selection - cannot share just one fact
- ✗ Documents can be stored indefinitely without consent
- ✗ Easy to correlate across verifiers via identical documents
- ✗ Forgery risk - verifier cannot cryptographically verify authenticity
- ✗ User has zero control after the document leaves their hands

FORTYTWO
SECURE TODAY. FUTURE READY.

Approach 2: Centralized Identity (OAuth / SSO)



PROBLEMS

- Single point of failure - one breach exposes ALL
- IdP sees ALL your activity across every service
- Provider can link all your sessions over time
- Trust concentration in one corporate entity
- IdP can be coerced by governments/courts
- Vendor lock-in & service discontinuation risk

FORTYTWO
SECURE TODAY. FUTURE READY.

Attack Scenario: The Correlation Attack

Priya presents the same credential to Bank (Day 1) and Insurance (Day 5)

1 Priya → presents credential to Bank

Bank records: session ID #A1, timestamp, credential fingerprint α

2 Priya → presents same credential to Insurance

Insurance records: session ID #B5, timestamp, credential fingerprint α

3 Bank & Insurance share session data

"Data partnership agreement" - both institutions, same parent group

4 Correlation found: fingerprint α appears in both!

Same credential hash used → definitively same user

5 Profiles merged across institutions

Bank now knows: Priya applied for insurance. Insurance knows: she has a loan.

6 PRIVACY COMPROMISED

Risk scoring without consent → higher premiums, denied loan, targeted ads

Root Cause: The credential fingerprint is IDENTICAL across presentations - making the user trivially linkable.

FORTYTWO
SECURE TODAY. FUTURE READY.

Attack: The Replay Attack

DEFINITION: An attacker captures a valid credential presentation and re-uses it later to impersonate the victim.

1	**Priya**	Generates credential presentation σ and sends it to Verifier A (e.g., Bank)
2	**Verifier A (Malicious)**	Accepts the proof... but also secretly SAVES a copy of σ for later use
3	**Attacker / Mal. Verifier**	Days later, replays σ to Verifier B (Insurance) - claiming to be Priya
4	**Verifier B (Honest)**	Cannot distinguish replay from genuine proof - ACCEPTS. Priya impersonated!

WHY IT WORKS: No binding to verifier or time!

FORTYTWO
SECURE TODAY. FUTURE READY.

Attack: Verifier Collusion

Scenario: Two 'honest-but-curious' verifiers follow the protocol but share data afterwards. Both belong to the same parent conglomerate.

BANK KNOWS:

- Income: ₹12 LPA
- Monthly spending pattern
- EMI obligations
- Loan applications history
- Credit card usage

+

INSURANCE KNOWS:

- Age: 24
- Non-smoker status
- Health conditions
- Family medical history
- Previous claims

COMBINED PROFILE (without Priya's knowledge or consent):

Full financial status + complete health history → automated risk scoring → higher premiums, denied loans, targeted manipulation

FORTYTWO
SECURE TODAY. FUTURE READY.

Known Attack Vectors - Taxonomy

ID	ATTACK NAME	DEFINITION	MECHANISM	SEVERITY
A1	**Linkability Attack**	Correlating user identity across multiple independent verification sessions	Verifiers compare credential signatures/hashes	HIGH
A2	**Replay Attack**	Reusing a previously captured valid credential presentation	No challenge binding / no session freshness	HIGH
A3	**Collusion Attack**	Multiple verifiers share data to reconstruct the holder's full profile	Data sharing agreements / same parent company	HIGH
A4	**Metadata Leakage**	Inferring identity through timing, IP address, device fingerprint	Side channels independent of credential content	MED
A5	**Attribute Correlation**	Inferring hidden attributes by combining revealed ones	Statistical inference / ML profiling	MED
A6	**Issuer Tracking**	The issuer monitors every time a credential it issued is presented	Issuer-verifier back-channel / credential serial numbers	MED

FORTYTWO
SECURE TODAY. FUTURE READY.

This Isn't Theoretical

Every major breach below used a 'trusted' centralized identity or data system:

Aadhaar / UIDAI

2023

81.5 Cr

Indian citizens' records

Names, addresses, phone numbers, Aadhaar numbers exposed via alleged government website vulnerability

Cambridge Analytica

2018

87M

Facebook profiles harvested

Personal data scraped via third-party app and used for targeted political advertising without consent

Equifax

2017

147M

consumer records exposed

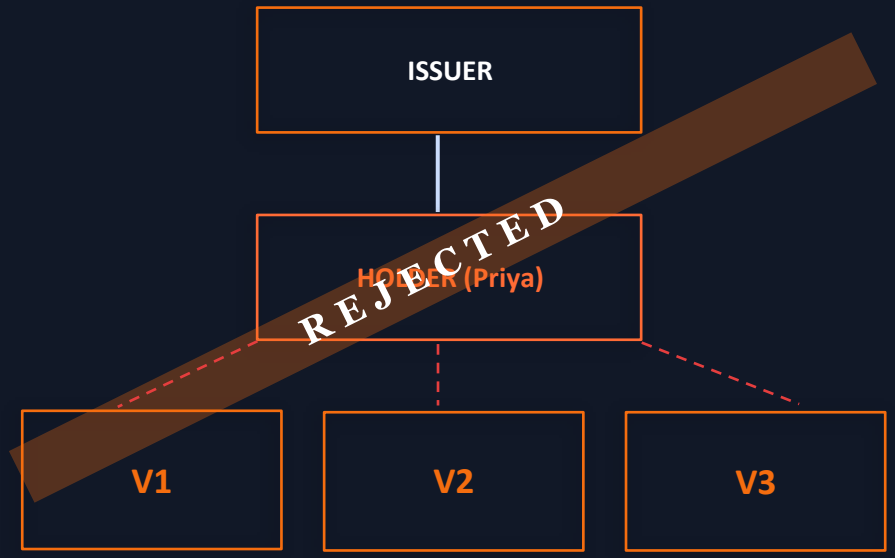
SSNs, birth dates, addresses, credit card data leaked - centralized credit bureau with no user control

In every case: users had no control over data after sharing. A technical privacy solution - not just policy - is needed.

FORTYTWO
SECURE TODAY. FUTURE READY.

Conventional Systems: VERDICT

THE TRADITIONAL ARCHITECTURE



KEY FAILURES

Cannot prevent over-disclosure

Verifier always gets more than needed

Cannot prevent linkability

Credential fingerprint enables correlation

Cannot survive collusion

Combined data creates full profile

Trust model is broken

Central parties become surveillance nodes

FORTYTWO
SECURE TODAY. FUTURE READY.

What Would a Proper Solution Look Like?

THE REQUIREMENTS

- Prove a FACT without revealing the underlying data
- Present to multiple verifiers without being LINKED
- Withstand honest-but-curious adversaries
- Remain private EVEN if verifiers collude

Section 3: Enter ZKP - The Solution that satisfies ALL four requirements →

ZKP

ENTER:

Zero-Knowledge Proofs

Conceptualized: 1985

SECTION 3 - ZERO-KNOWLEDGE PROOFS

Enter ZKP: The Solution

3

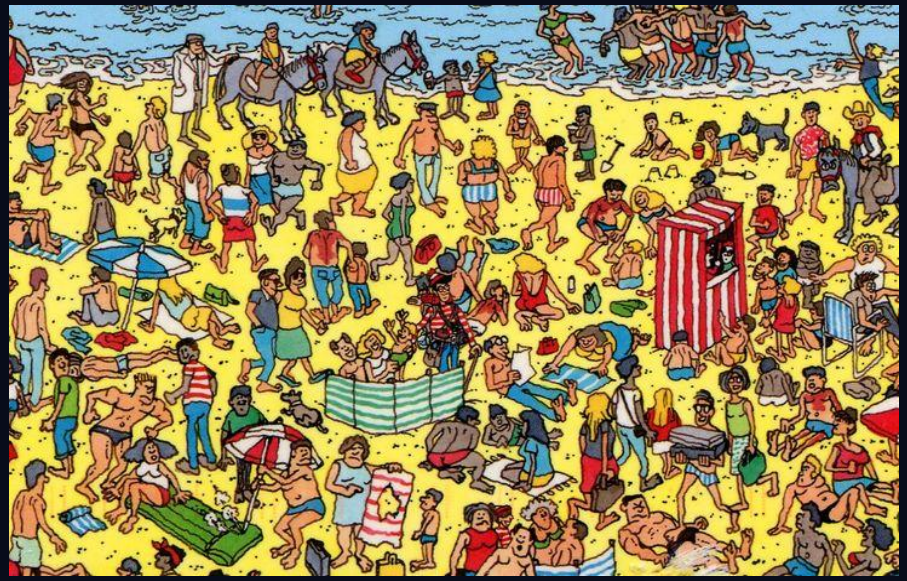
"Prove everything. Reveal nothing."

Slides 23–35

FORTYTWO
SECURE TODAY. FUTURE READY.

Where is Waldo?

Peggy knows the location of Waldo. She wants to PROVE she knows it without say.



FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP: Formal Definition & Three Properties

Goldwasser, Micali & Rackoff - MIT 1985. Turing Award 2012. Foundation of all modern privacy-preserving cryptography.

DEFINITION

A ZKP is a protocol between Prover (P) and Verifier (V) whereby P proves a statement x ∈ L is TRUE without revealing any information beyond the truth of that statement.

COMPLETENESS

If the statement is TRUE, an honest prover can always convince an honest verifier

PRIYA: Priya's income IS ₹12 LPA → she generates a valid proof → bank always accepts

$$\Pr[V \text{ accepts } P(x)] = 1$$

SOUNDNESS

If the statement is FALSE, no cheating prover can convince the verifier (except negligible probability)

PRIYA: Ravi's income = ₹3 LPA → cannot forge valid proof → bank always rejects

$$\Pr[V \text{ accepts } P^*(x)] \leq \varepsilon$$

ZERO-KNOWLEDGE

The verifier learns NOTHING beyond the yes/no answer - the transcript can be simulated without the witness

PRIYA: Bank verifies income > 5LPA → learns only 'true' → salary ₹12 LPA never revealed

$$\text{View}_V(P,x) \equiv \text{Sim}(x)$$

FORTYTWO
SECURE TODAY. FUTURE READY.

Parties in a ZKP System

ISSUER

Trusted Authority

Signs user attributes. Does NOT participate in individual proof sessions.

σ = Sign(sk_ISI, attrs)

One-time issuance, offline

e.g. Aadhaar Authority, ISI

Signs: "Priya income=₹12LPA" with its master key

PROVER / HOLDER

Priya (The User)

Holds the credential. Generates a ZKP to prove a specific predicate without revealing the underlying value.

π = Prove(pk, witness, x)

Chooses WHICH predicate to prove

Controls entire disclosure

Proves: income > ₹5LPA without revealing ₹12LPA

VERIFIER

Bank / Insurance / Portal

Receives the ZKP and verifies its mathematical validity. Gets only a YES/NO - never raw data.

b = Verify(vk, π, x) → {0,1}

Cannot extract attribute value

Cannot link across sessions

Checks proof validity → grants loan ✓

Priya Flow: ISSUER signs credential once → PROVER generates selective proof per session → VERIFIER checks proof only

FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP System Flow - Five Phases

1 SETUP	2 ISSUE	3 PROVE	4 TRANSMIT	5 VERIFY
Trusted Parameters	Credential Issuance	Proof Generation	Send Proof Only	Proof Verification
Generate CRS / SRS	ISI signs attributes	Select predicate p	Send only π to verifier	b = Verify(vk, π, x)
Proving key (pk)	σ = Sign(sk, attrs)	π = Prove(pk, w, x)	~200 bytes typical	Outputs: {0, 1}
Verif. key (vk)	Priya stores σ	w = secret witness	No raw data	No witness revealed
Done once globally	Offline - happens once	Runs on Priya's device	Session-nonce bound	Priya stays unlinkable

Priya participates in phases 2–4 only. Verifier never sees her witness. Issuer not involved in phases 3–5.

FORTYTWO
SECURE TODAY. FUTURE READY.

Key Concepts: Witness, Statement & Circuit

WITNESS (w)

`w ∈ Witness(L, x)`

The SECRET data the prover knows but does NOT reveal

EXAMPLE:

w = Priya's income ₹12 LPA
The Aadhaar number, age, credential σ

Private - only prover knows w

Used internally in Prove()

Never transmitted to verifier

STATEMENT (x)

`x ∈ L (language membership)`

The PUBLIC CLAIM the prover wants to prove is true

EX:

x = "income > ₹5 LPA"
"age ≥ 18", "has M.Tech degree"

Public - shared with verifier

Boolean: true or false

The ONLY thing verifier learns

CIRCUIT (C)

`C(w, x) = 1 iff claim holds`

The COMPUTATION relating witness to statement, encoded as arithmetic gates

EX:

C(w=12, x=5) → 12>5 → 1
Encoded as R1CS / Plonkish constraints

Defines the allowed check

Fixed per predicate type

Compiled once, verified many times

`Core equation: π = Prove(pk, w, x) | b = Verify(vk, π, x) | C(w, x) = b | Verifier knows only x and b - never w`

FORTYTWO
SECURE TODAY. FUTURE READY.

Interactive vs. Non-Interactive ZKPs

INTERACTIVE ZKP

Multiple rounds of challenge / response

Requires both parties ONLINE simultaneously

Does not scale to async internet use

Simpler to analyze formally

VS

NON-INTERACTIVE ZKP (NIZK)

Single message. No back-and-forth.

Fiat–Shamir Heuristic: replace challenge with H(commitment)

Setup: Public CRS/SRS generated once (trusted setup or transparent)

Priya: $\pi = \text{Prove}(pk, w, x) \rightarrow$ single artifact, no interaction

Bank: $b = \text{Verify}(vk, \pi, x) \rightarrow \{0,1\}$ async, offline OK

No simultaneous online requirement

Scales to web and mobile

Basis: zk-SNARKs, zk-STARKs, Bulletproofs

The hackathon system will use NON-INTERACTIVE ZKPs - practical, async, single proof artifact per verification session.

FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP vs. Traditional: Head-to-Head

REQUIREMENT	TRADITIONAL	ZKP SYSTEM
Data shared with verifier	✗ Full credential / document	✓ Proof π only (~200 bytes)
Linkability across sessions	✗ Same ID → trivially linkable	✓ Fresh random proof each time
Collusion survivability	✗ Profiles merge easily	✓ No data to merge - only bits
Attribute selectivity	✗ All-or-nothing disclosure	✓ Prove any predicate you choose
Verifier trust requirement	✗ Must trust verifier's storage	✓ No trust needed - math verifies
Replay protection	✗ Credential can be replayed	✓ Nonce binding prevents replay
Issuer involvement per use	✗ Often required each time	✓ One-time issuance, infinite proofs

FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP Solves Priya's Four Problems

1 OVER-DISCLOSURE

BEFORE

Bank gets full salary slips + bank statements + PAN = ~19 MB

AFTER ✓

Bank gets π (proof: income > ₹5 LPA) = ~200 bytes

🏆 **99.999% data reduction - by construction**

2 LINKABILITY

BEFORE

Same credential fingerprint α links Priya's Bank and Insurance sessions

AFTER ✓

Each ZKP uses fresh randomness - π_1 and π_2 are computationally unlinkable

🏆 **Simulation-based unlinkability - proven**

3 COLLUSION

BEFORE

Bank + Insurance combine data → full Priya profile assembled

AFTER ✓

No data was shared → collusion yields only {true, true} - useless bits

🏆 **Collusion-resistant by construction**

4 LOSS OF CONTROL

BEFORE

Once document shared, Priya has zero control over it forever

AFTER ✓

Priya generates a fresh selective proof per session - no persistent data leaves

🏆 **Holder-controlled, technically enforced**

FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP Walkthrough: Priya's Bank Loan

SETUP — **ISI Authority**

Generate public parameters: CRS, proving key pk, verification key vk

Done once globally - all parties use the same params

ISSUANCE — **ISI → Priya**

ISI signs Priya's attributes: σ = Sign(sk_ISI, age=24, income=12LPA, edu=MTech)

Offline step - Priya stores σ securely, never shares it

REQUEST — **Bank → Priya**

Bank publishes predicate: 'Prove income > ₹5 LPA' with challenge nonce η = 0xFA3B

Bank sends (predicate, η, vk). No raw data requested from Priya

PROVE — **Priya's Wallet**

Priya computes: π = Prove(pk, w=(income=12, σ), x=(income>5, η))

Circuit: 12 > 5 = TRUE. Proof generated locally - nothing leaves device

VERIFY — **Priya → Bank**

Priya sends ONLY π. Bank runs b = Verify(vk, π, x) → output: b=1 (VALID)

Bank grants loan. Income ₹12 LPA was NEVER transmitted at any point

REPEAT — **Insurance, Employer...**

Priya generates fresh π_2 (age ≥ 18) for Insurance, π_3 (M.Tech) for Employer

π_1, π_2, π_3 are computationally unlinkable - no cross-verifier tracking

FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP Security Model & Trust Assumptions

What ZKPs provably protect against - and what they still assume:

ADVERSARIES ZKP HANDLES

Honest-but-Curious Verifier	Tries to extract witness from π - impossible by ZK property
Malicious Prover (no witness)	Cannot produce valid proof - blocked by soundness
Colluding Verifiers	Combine outputs: just {1, 1} - no useful info
Replay Attacker	Proof bound to session nonce η - replay = invalid
Correlation / Linkage Attack	Fresh randomness per proof - unlinkable by simulation

TRUST ASSUMPTIONS

- Trusted Setup: CRS generated honestly (MPC ceremonies mitigate this)
- Hardness: Security relies on computational hardness assumptions (DL, pairing)
- Issuer honesty: Issuer correctly signs real attributes (policy, not ZKP)
- Prover's device: Witness must remain secure on Priya's device

⚠ WHAT ZKP DOES NOT SOLVE (requires system design beyond ZKP):

Does NOT prevent verifier from acting on the result (e.g., denying a loan) - requires policy/law.
Does NOT guarantee issuer signed correct attributes - requires trust in issuer.
Does NOT prevent metadata leakage (IP address, timing) - requires network-layer anonymity (Tor, mixnets).

FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP Properties - Applied Checklist

✓ Proves knowledge without revealing underlying data	*Zero-Knowledge property*	*Goldwasser–Micali–Rackoff 1985*
✓ Honest prover always convinces verifier	*Completeness*	*Pr[V accepts] = 1*
✓ Cheating prover cannot produce valid proof	*Soundness*	*Pr[V accepts false] ≤ ε*
✓ Each proof unlinkable across different verifiers	*Simulation-based unlinkability*	*Fresh randomness*
✓ Non-interactive - single proof artifact	*NIZK via Fiat-Shamir*	*Random oracle model*
✓ Selective disclosure of individual attributes	*Predicate proofs per attribute*	*Circuit per predicate*
✓ Withstands honest-but-curious adversaries	*Computational ZK*	*View simulatability*
? Prevents verifier acting on the verified result	*(policy layer - out of scope)*	*Legal / contractual*

ZKP SCORE

7/7

requirements fully met

(+1 policy note)

0 failing criteria

FORTYTWO
SECURE TODAY. FUTURE READY.

Section 3 Summary: Key Takeaways

1 ZKP EXISTS - and it works

Goldwasser–Micali–Rackoff 1985 proved these systems exist. Modern NIZK systems (SNARKs/STARKs) have proof generation times of milliseconds.

2 It Satisfies ALL Three Core Properties

Completeness, Soundness, and Zero-Knowledge together give us exactly what the identity problem requires - nothing more, nothing less.

3 It Directly Addresses Priya's Four Problems

Over-disclosure → 200 bytes. Linkability → fresh proofs. Collusion → no data shared. Loss of control → holder generates proof herself.

4 It Requires Careful System Design

ZKP is not magic. Trusted setup, issuer correctness, metadata channels, and policy enforcement still require thoughtful engineering.

NEXT → Section 4: ZKP Protocols Landscape - zk-SNARKs, zk-STARKs, Bulletproofs. Which scheme fits Priya's hackathon challenge?

FORTYTWO
SECURE TODAY. FUTURE READY.

SECTION 4 - PROTOCOLS & PRIMITIVES

ZKP Protocols Landscape

4

"Not all zero-knowledge proofs are created equal."

Slides 36–38

FORTYTWO
SECURE TODAY. FUTURE READY.

Major ZKP Protocols at a Glance

PROTOCOL	TYPE	TRUSTED SETUP	PROOF SIZE	VERIFY SPEED	PQ-SECURE	BEST FOR
Groth16 2016	zk-SNARK	YES	~192 bytes	~1 ms	NO	Ethereum L2, Zcash (Sapling)
PLONK 2019	zk-SNARK	Universal	~500 bytes	~5 ms	NO	General circuits; Aztec, zkSync
zk-STARKs 2018	zk-STARK	NONE	10–200 KB	~50 ms	YES	StarkNet, StarkEx, post-quantum
BBS+ 2023	Signature	CRS only	~112 bytes	<1 ms	NO	W3C VC, multi-attr selective disclosure
Bulletproofs 2017	NIZK Range	NONE	~1 KB	~100 ms	NO	Monero (range proofs), confidential txns

HACKATHON GUIDANCE: For multi-attribute identity credentials, BBS+ or PLONK are strong starting points. For pure range proofs, consider Groth16.

FORTYTWO
SECURE TODAY. FUTURE READY.

Choosing Your Protocol

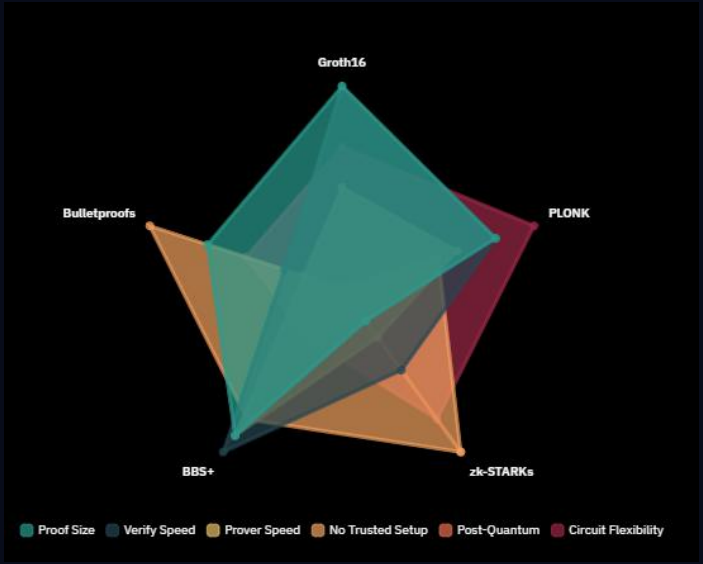
Use this decision guide for your hackathon submission:

DECISION FLOWCHART

What's your PRIMARY constraint?

- **Proof Size** → **GROTH16 or BBS+**
- **No Trusted Setup** → **zk-STARKs or Bulletproofs**
  - Need UNLINKABILITY? --> BBS+ with fresh blinding per proof
- **Credential Flexibility** → **BBS+ or PLONK**

For THIS hackathon: **BBS+ (attribute proofs) + Groth16 (range proofs) + PLONK (general predicates)**



FORTYTWO
SECURE TODAY. FUTURE READY.

SECTION 5 - REAL-WORLD DEPLOYMENTS

ZKP in the Real World

5

"From theory to production - who is using ZKP today?"

Slides 39–46

FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP is Already in Production

These are live deployments - not prototypes:

Consumer | BBS+ / W3C VC

Google Wallet Age Verification

Uses ZKP to prove age ≥ 18 from digital driving licence without revealing DOB, name, or address.

USA, select states - live 2023

Blockchain | Groth16 / zk-SNARK

Zcash - Shielded Transactions

Proves transaction validity (inputs = outputs + fees) without revealing sender, receiver, or amount.

Global - live since 2016

Scaling | PLONK / STARK

Polygon zkEVM / StarkNet

Proves Ethereum computation off-chain using ZKP. Enables 100× scalability with full Ethereum security.

Global mainnet - live 2023

Identity | SD-JWT / BBS+

EU Digital Identity Wallet (EUDI)

EU regulation mandating selective disclosure of identity attributes. ZKP used for unlinkable presentations.

EU - pilot 2024, mandatory 2026

Governance | zk-SNARK / Groth16

MACI - Minimal Anti-Collusion

Proves vote was cast and counted correctly without revealing the vote. Used in Gitcoin, ETH governance.

Global - DeFi governance

Finance | Custom ZKP / BBS+

ING / JP Morgan ZK Credit

Proves credit score above threshold or AML compliance without sharing full financial records with counterparty.

Enterprise - pilot / production

FORTYTWO
SECURE TODAY. FUTURE READY.

Case Study: Google Wallet Age Verification

HOW IT WORKS - STEP BY STEP

1. **DMV issues credential**
   State DMV issues a W3C Verifiable Credential containing: Name, DOB, address, licence class, photo hash

2. **User stores in Google Wallet**
   Credential stored as BBS+ signed document. Private key stays on device - never leaves.

3. **Website requests age verification**
   Site sends a Presentation Request: 'Prove age ≥ 18 - nothing else needed'

4. **Wallet generates ZK proof**
   BBS+ derives a selective disclosure proof π that commits to DOB without revealing it. Includes fresh random blinding factor.

5. **Site verifies proof**
   Site checks proof π against DMV public key. Learns only: 'User is ≥ 18.' Cannot learn DOB, name, address - ever.

PROPERTIES ACHIEVED

- **Selective Disclosure**
  Only 'age ≥ 18' revealed - not DOB
- **Unlinkability**
  Fresh blinding factor per presentation - websites can't correlate
- **No Raw Data Shared**
  Actual credential bytes never leave the wallet
- **Issuer Not Notified**
  DMV is NOT pinged per verification - offline proof
- **Cryptographic Binding**
  Only holder of the private key can generate valid proof

Standards: W3C VC Data Model 2.0 · DIF BBS+ Draft · ISO 18013-7 (mDL)

FORTYTWO
SECURE TODAY. FUTURE READY.

Case Study: Zcash Shielded Transactions

STATEMENT BEING PROVEN:

"I know a valid spending key, the note commitment opens correctly, and inputs = outputs + fee - without revealing sender, receiver, or amount"

TRANSPARENT TX (Bitcoin-style)	SHIELDED TX (Zcash Groth16)
✗ Sender address: 1A2B3C...XYZ	✓ Sender: cryptographic commitment only
✗ Receiver address: 4D5E6F...ABC	✓ Receiver: encrypted note - only they can open
✗ Amount: 2.5 ZEC	✓ Amount: hidden (pedersen commitment)
✗ Transaction graph: fully traceable	✓ Validity: ZK proof π of conservation
✗ History: every hop forever visible	✓ History: graph is completely opaque

GROTH16 CIRCUIT STATS (Zcash Sapling):

Proof Size	Verify Time	Circuit Size	Prover Time	Trusted Setup
~192 bytes	~1.4 ms	~2M gates	~6 seconds	Sapling MPC

FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP for KYC / AML Compliance

Banks are legally required to verify customers - but does that mean sharing all data with every counterparty?

CURRENT PAIN: Every bank, exchange, and counterparty runs its own KYC. Same customer submits passport + proof of address + bank statements to 10+ institutions - full re-disclosure each time.

Regulated KYC Provider

(e.g., Sumsub, Onfido)

Conducts full KYC once. Issues signed credential: 'PAN verified, no PEP, no sanctions'

Priya (Customer)

holds credential on device

Generates ZK proof per institution. Proof attests compliance - no raw documents shared

Bank / Exchange

(verifier)

Verifies proof in <5ms. Knows customer is KYC-compliant. Does NOT receive underlying documents

WHAT ZKP KYC ENABLES:

- **KYC once, reuse everywhere**
  Customer never re-submits raw documents
- **Minimal data per institution**
  Each bank learns only: 'compliant / not compliant'
- **Unlinkable sessions**
  Bank A cannot tell Bank B they share the same customer
- **Revocable credentials**
  KYC provider can revoke if status changes - proof fails automatically

FORTYTWO
SECURE TODAY. FUTURE READY.

Standards: Where ZKP Is Being Codified

W3C — World Wide Web Consortium

VC Data Model 2.0 — Recommendation
Defines structure of Verifiable Credentials and Presentations

BBS Cryptosuite — Draft
ZK selective disclosure for VCs using BBS+ signatures

vc-di-ecdsa — Candidate Rec.
Elliptic curve data integrity proofs for VCs

DIF Presentation Exchange — Community
Query language for requesting specific VC attributes

IETF — Internet Engineering Task Force

SD-JWT (RFC 9421) — Proposed Std
Selective disclosure using hashed salted claims in JWT

SD-JWT VC Profile — Draft
VC profile for SD-JWT - used in EU EUDI wallet

OAuth 2.0 VC Issuance — Draft
OAuth flows adapted for credential issuance & presentation

BBS IETF Draft — Draft
BBS signature scheme under IETF standardization

ISO / IEC — International Standards Organization

ISO 18013-5 (mDL) — Published
Mobile Driving Licence standard - used by Google/Apple Wallet

ISO 18013-7 — Published
Online mDL presentation - ZKP selective disclosure variant

ISO 23220 - MDOC — Under dev.
Generic mobile document framework beyond driving licences

ISO/IEC 24760 IAM — Published
Identity management framework including credential systems

Your hackathon solution should align with at least one standard (recommended: W3C VC + BBS+ Cryptosuite or SD-JWT)

FORTYTWO
SECURE TODAY. FUTURE READY.

ZKP Adoption Maturity by Domain

DOMAIN	Research	Pilot	Production	At Scale	EXAMPLES
Blockchain / DeFi			✓	✓	Zcash, Tornado Cash, zkSync
L2 Scaling (zkRollups)			✓	✓	StarkNet, Polygon zkEVM, zkSync Era
Digital Identity (mDL/VC)		✓	✓		Google Wallet, EU EUDI Wallet
KYC / AML Compliance		✓	✓		ING, JP Morgan pilots
Healthcare Records	✓	✓			Research prototypes; NIH pilots
Voting / Governance		✓	✓		MACI - Gitcoin, ETH governance
Supply Chain Provenance	✓	✓			IBM Food Trust, Walmart pilots
ZK Email / Web2→Web3 Bridge	✓	✓			zk-email.org, Proof of Twitter

FORTYTWO
SECURE TODAY. FUTURE READY.

Real-World Lessons for Your Design

01 ZKP is production-ready

Zcash (2016), StarkNet (2022), Google Wallet (2023) - ZKP is not academic. Real-world deployments handle millions of transactions.

Lesson: Your hackathon solution should target production-grade properties, not just proof-of-concept.

02 Standards convergence is happening

W3C VC 2.0 + BBS+ Cryptosuite + SD-JWT are converging into a unified credential ecosystem. EUDI wallet mandates these by 2026.

Lesson: Align to W3C VC Data Model - your solution becomes interoperable with Google Wallet, EUDI, and more.

03 Selective disclosure > full disclosure

Every production system studied reveals minimum data. Google Wallet doesn't reveal DOB. Zcash doesn't reveal amounts. BBS+ enables this natively.

Lesson: Design your credential schema for selective attribute disclosure from day one.

04 Unlinkability requires fresh randomness

The critical difference in BBS+: each presentation uses a fresh blinding factor, making proofs to different verifiers computationally unlinkable.

Lesson: Your multi-verifier architecture must generate independent, re-randomized proofs per presentation.

Next: Section 6 - Bridging all of this to your Hackathon submission →

SECTION 6 - HACKATHON BRIDGE

Bridging to Your Hackathon

6

"Everything you learned today - applied to a real problem."

Slides 47–50

FORTYTWO
SECURE TODAY. FUTURE READY.

Your Challenge: Privacy-Preserving Identity

THE MISSION

Design and implement a ZKP-based identity verification system where Priya can prove specific attributes to multiple verifiers without disclosing raw credential data - and without any two verifiers being able to link her presentations.

WHO ARE THE ACTORS?

- Issuer: UIDAI / Govt. authority - issues Aadhaar-backed VC
- Holder: Priya - stores credential, generates ZK proofs
- Verifier 1: IndusInd Bank - needs income > ₹5 LPA
- Verifier 2: LIC Insurance - needs age ≥ 18, non-smoker
- Verifier 3: Zomato - needs age ≥ 21 for alcohol delivery

WHAT MUST BE PROVEN?

- Income > ₹5,00,000 without revealing exact salary
- Age ≥ 18 (or ≥ 21) without revealing DOB
- Non-smoker status - without health records
- Valid govt. ID - without Aadhaar number
- Address in India - without revealing PIN code

HARD CONSTRAINTS

- No raw credential data may be transmitted
- No two verifiers can correlate presentations
- Issuer must NOT be contacted per verification
- System must handle credential revocation
- Must align with W3C VC Data Model 2.0

SUCCESS = All four ZKP properties hold · All attributes proven · Zero raw data exposed · Fully unlinkable across verifiers

FORTYTWO
SECURE TODAY. FUTURE READY.

Reference Architecture for Your Solution

LAYER 1 - CREDENTIAL ISSUANCE

Wallet App
(iOS/Android)

UIDAI
(Issuer)

BBS+ signed
VC (W3C 2.0)

Priya
(Holder)

LAYER 2 - ZKP PROOF GENERATION

Select Attributes → **Compute Blinding** → **Generate Proof π** → **Re-randomize per V**

LAYER 3 - MULTI-VERIFIER PRESENTATION

Bank
income>5L
π_1

Insurance
age≥18,NS
π_2

Zomato
age≥21
π_3

Layer 4 - OFF-CHAIN VERIFICATION + REVOCATION (Status List 2021 / RevReg)

RECOMMENDED TECH STACK

Credential Format
W3C VC Data Model 2.0 · @digitalbazaar/vc

ZKP Scheme
BBS+ Signatures · @mattrglobal/bbs-signatures

Predicate Proofs
Groth16 / PLONK · snarkjs or circom

Wallet SDK
DIF Universal Wallet · @sphereon/ssi-sdk

Presentation Req.
DIF Pres. Exchange · @sphereon/pe-js

Revocation
Status List 2021 · @digitalbazaar/vc-revocation-list

Testing
Jest + Hardhat (EVM) · hardhat + waffle

github.com/42hackathon-2026/zkp-identity-starter

FORTYTWO
SECURE TODAY. FUTURE READY.

How You Will Be Judged

CRITERION	WEIGHT	FULL MARKS (what earns them)
ZKP Properties (With proper documentation)	30%	All four properties (Completeness, Soundness, ZK, Unlinkability) formally argued and demonstrated
Multi-Verifier Unlinkability	20%	Proof transcripts for Bank, Insurance, and Zomato are computationally unlinkable - demonstrated
Implementation (With a proper sequential Video Presentation)	30%	Working code, clean repo, W3C-compatible credential schema, integration tests passing
Innovation / Bonus	20%	PQC Variant, Revocation handling, cross-chain anchoring, zk-email binding, or novel circuit design

BUILD IT. PROVE IT. WIN IT.

Q&A

Feel free to ask about the problem, demo, architecture, security, roadmap, or impact if any.

STATISTICAL INDIAN INSTITUTE

UNITY IN DIVERSITY

FORTYTWO

SECURE TODAY. FUTURE READY.